

# VICTOR BARRETO ANDRADE GONÇALVES

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## PROFESSIONAL SUMMARY

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Computing Engineering student with hands-on experience in High Performance Computing, machine learning, and autonomous systems. Proven track record in applied AI projects with real-world industrial impact, including computer vision for automated inspection and neural network models for predictive analysis. Seeking to contribute to Ford's innovation and manufacturing excellence through data-driven solutions, process optimization, and emerging technologies.

## EDUCATION

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**Bachelor's Degree – Computing Engineering** – SENAI CIMATEC University, Salvador, BA *Jan 2024 – Dec 2028 (expected)*

**Bachelor's Degree – Information Systems** – Universidade Federal da Bahia, Salvador, BA *2026 – 2030 (expected)*

## HONORS & AWARDS

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- 6th Place – AWS DeepRacer National Final (CSBC 2025) · Autonomous vehicle AI racing using reinforcement learning · 3rd Place Northeast Semifinal · Top 10 among all Brazilian universities, representing CIMATEC and the state of Bahia.
- 2nd Place – Scientific Article Competition, Seminário Desafio Pesquisador, SENAI CIMATEC.

## PROFESSIONAL EXPERIENCE

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**HPC Intern – High Performance Computing** – SENAI CIMATEC, Salvador *Oct 2025 – Present*

- Developed and deployed high-performance computational solutions for molecular simulation and drug discovery, applying deep learning models (PyTorch, TensorFlow) to accelerate predictive pharmaceutical analysis.
- Optimized deep learning pipelines in cluster-based HPC environments using SLURM, achieving significant reductions in computation time through process parallelization and efficient GPU resource allocation.
- Gained hands-on experience with large-scale data processing and simulation workflows directly applicable to industrial engineering and manufacturing R&D environments.

## LEADERSHIP & VOLUNTEER EXPERIENCE

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**AWS Cloud Club Captain** – AWS Cloud Club – SENAI CIMATEC, Salvador *Nov 2025 – Present*

- Selected by Amazon Web Services to lead and grow the campus cloud computing community; organized workshops, hackathons, and educational events on AWS technologies for 100+ students.
- Promoted cloud-based innovation and collaborative problem-solving, fostering a culture of technical excellence aligned with industry 4.0 demands.

**Treasurer & Project Advisor** – IEEE Robotics and Automation Society – Student Chapter, Salvador *Mar 2024 – Present*

- Project Advisor (2025–2026): Led a multidisciplinary team developing a CNN-based computer vision system for automated X-ray analysis (pneumonia and tuberculosis detection), achieving high diagnostic accuracy. Guided data science best practices, ethical AI in healthcare, and clinical interface design.

- Treasurer (Jan 2025–present): Managed strategic financial planning, budget control, and resource mobilization through IEEE grants and external sponsorships, ensuring chapter financial sustainability.
- Trainee (2024): Co-developed Leak Track — an autonomous inspection robot integrating ROS2, CNNs, and FLIR thermal imaging for real-time leak detection in restricted-access industrial structures, directly relevant to manufacturing and plant safety.

**Volunteer Teaching Monitor – Competitive Programming – SENAI CIMATEC Programming Club, Salvador** *Nov 2024 – Nov 2025*

- Taught algorithms, data structures, and optimization techniques in C++ to university students, creating exercise sets based on the Brazilian Olympiad in Informatics (OBI) and the SBC Programming Marathon.

## SKILLS

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**Programming Languages:** Python, C++

**Frameworks & Tools:** PyTorch, TensorFlow, Keras, SLURM, Git, GitHub, GitLab

**Technical Areas:** High Performance Computing (HPC), Machine Learning & Deep Learning, Computer Vision, Autonomous Systems (ROS2), Cloud Computing (AWS), Bioinformatics, Competitive Programming

**Languages:** Portuguese (Native), English (C2 – EF SET Proficient)

## INTERESTS

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Industrial automation and manufacturing intelligence · High Performance Computing · AI applied to process optimization · Autonomous systems and robotics · Cloud Computing · Competitive Programming · Education through technology